



UNIVERSITY OF RUHUNA
BACHELOR OF INFORMATION AND COMMUNICATION TECHNOLOGY
ICT4163 - Digital Image Processing
Practical 04 – Intensity Transformation



Note that: Download the given resources from the Resources Folder 2 under the lesson 05.

1.
 - a. Define a Python function named `negative` that accepts one input parameter `r`.
 - b. Inside the function, compute the negative intensity value using the formula:
$$s = 255 - r$$
 - c. return the output value `s` from the function.
2.
 - a. Use NumPy to generate 1000 evenly spaced intensity values between 0 and 255.
 - b. Store the values in a variable named `x_values`.
 - c. Apply the `negative()` function to the generated input values.
 - d. Store the output values in a variable named `y_values`.
 - e. Use **Matplotlib** to **plot the input intensity values against the output intensity values**.
 - f. Properly label the x and y axis of the plot.
3.
 - a. Write the Python statement to **read the image file** named: "ship.jpeg"
 - b. Store the image in a variable named **image**.
 - c. Apply the negative transformation to the image stored in **image**.
 - d. store the result in a variable named **image_negative**.
 - e. Visualize original image and transformed image images using **Matplotlib**.
4.
 - a. Write the Python statement to **define a function named scale_image** that accepts one parameter called **input_img** that converts the input image intensity values to the Range 0–255.
 - b. Define a Python function named `logtransform` that accepts an input parameter `r` and a parameter `c` with a default value 1.
 - c. Inside the function, compute the log intensity transformation value using the formula:
$$s = c \log(1 + r)$$

- d. Return the output value s from the function.
- 5.
- a. Use NumPy to generate 1000 evenly spaced intensity values between 0 and 255.
 - b. Store the values in a variable named `x_values`.
 - c. Apply the `logtransform()` function to the generated input values.
 - d. Store the output values in a variable named `y_values`.
 - e. Use **Matplotlib** to **plot the input intensity values against the output intensity values**.
 - f. Properly label the x and y axis of the plot.
- 6.
- a. Write the Python statement to **read the image file** named: "road.jfif"
 - b. Store the image in a variable named **roadimage**.
 - c. Apply the logtransformation to the image stored in **roadimage**
 - d. Store the result in a variable named **roadimagelog**.
 - e. Apply the scaling function to the **roadimagelog**.
 - f. Store the result in a variable named **scaledroadimagelog**.
 - g. Visualize original image and transformed image using **Matplotlib**.
- 7.
- a. Implement a Python function named `logtransformation`, and performed the log transformation with the given image named "brain.jfif". and implement another function named `inverse_log_transformation` and use the output of the `logtransformation` function as the input to the inverse log transformation function, and finally display the original image , log transformed image and the inverse log transformed image.
 - b. Use NumPy to generate 500 evenly spaced intensity values between 0 and 255.
 - c. Store the values in a variable named `x_values`.
 - d. Apply the `logtransform()` function to the generated input values.
 - e. Store the output values in a variable named `y_log_values`.
 - f. Apply the `inverse_logtransform()` function to the generated input values.
 - g. Store the output values in a variable named `y_inverse`.
 - h. **Use Matplotlib to plot the input intensity values against the output intensity values to display log transformation and inverse log transformation respectively.**

- i. Properly label the x and y axis of the plot.
8. Find two suitable images and perform the gamma transformation for the following gamma values.

Note: Every image we cannot use for gamma transformation. Before you answer this question study properly the images we use for gamma transformation.

Gamma values:

Gamma value set A: 0.2,0.4,0.6, - apply these values only for one image.

Gamma value set B: 2.0,4.0,6.0 - apply these values only for one image.

Compare the resultant images you obtained with respect to the each gamma value, and write 02 conclusions highlighting which gamma value became the best for each image.

9. Implement a piecewise linear transformation function that Maps input intensities (0–255) to output intensities $((0,0) \rightarrow (r1,s1), (r1,s1) \rightarrow (r2,s2), (r2,s2) \rightarrow (255,255))$ using the following 3 conditions. The function should accept 05 parameters including r,r1,s1,r2 and s2. r = input image, for r1, s1, r2 and s2 define suitable values based on the image you used.(you can use your own image as the input image)

Conditions for function implementation

When $r < r1$

$$s = (s1 / r1) * r$$

When $r > r1$ and $r < r2$:

$$s = ((s2 - s1) / (r2 - r1)) * (r - r1) + s1$$

Otherwise

$$s = ((255 - s2) / (255 - r2)) * (r - r2) + s2$$

- a. Converts the function you created to accept arrays.
 - b. Display the original image and the piecewiseLineartransformed image.
 - c. Use NumPy to generate 1000 evenly spaced intensity values between 0 and 255.
 - d. Store the values in a variable named x_values.
 - e. Apply the piecewiseLineartransformation function to the generated input values.
 - f. Store the output values in a variable named y_values.
 - g. Use Matplotlib to plot the input intensity values against the output intensity values.
 - h. Properly label the x and y axis of the plot.
10. Implement an intensity level transformation function based on the following condition.

Condition: If r between 120 and 200 assign 255 as the intensity, otherwise assign 0.

- a. Use a suitable image and apply the intensity level transformation function and display the original image and the transformed image.
- b. Use NumPy to generate 1000 evenly spaced intensity values between 0 and 255.
- c. Store the values in a variable named x_values.

- d. Apply the intensity level transformation function to the generated input values.
- e. Store the output values in a variable named `y_values`.
- f. Use Matplotlib to plot the input intensity values against the output intensity values.
- g. Properly label the x and y axis of the plot.